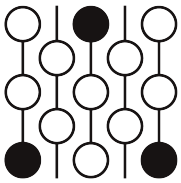
Review of Problem 6 – Astra

First Proof Third Batch
Reviewer 1
October 8–9, 2026



1 Review

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- −0.5: major errors
- −1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score:1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- −1: very poor exposition

Score:1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- −1: blatant plagiarism or hallucinations

Score:0

> **Reviewer 1**
> Overall we find that the exposition and correctness are worthy of their scores. We take off points for missing attributions for the Poincaré conjecture and not citing relevant literature for Lemma 6.

2 Submission with inline comments

Abstract

Every Dehn surgery on an L-space knot in S^3 has at most two nontrivial prime summands. We prove this using established results about reducing slopes and prime decompositions, together with a group-theoretic genus bound for integer homology sphere summands of surgeries on fibered knots.

3 Statement and conventions

We use the standard ambient-manifold convention: the knots under consideration lie in S^3. A rational homology sphere Y is an L-space if

$$\dim_{\mathbb{F}_2} \widehat{HF}(Y;\mathbb{F}_2) = |H_1(Y;\mathbb{Z})|.$$

An L-space knot is a knot admitting a nonmeridional L-space surgery. We allow either sign for this surgery coefficient; when the coefficient can be chosen positive, we explicitly say that the knot admits a positive L-space surgery.

> **Reviewer 1**
>
> Immediately in the proof of the main theorem they say you can always pick the surgery to be positive. Just say this here.

For a knot K, write $S^3_{p/q}(K)$ for Dehn filling along $p\mu + q\lambda$, where μ is a meridian and λ is the Seifert longitude. For finite slopes we take $\gcd(p,q) = 1$ and $q > 0$; the meridional slope is denoted by ∞.

All homology groups without specified coefficients have coefficients in $\mathbb{Z}$. Prime summands mean the non-S^3 factors in the Kneser–Milnor prime decomposition [Mil62, Theorem 1]. In particular, S^3 contributes no nontrivial prime summands.

Theorem 1. *Let $K \subset S^3$ be an L-space knot. For every $r \in \mathbb{Q} \cup \{\infty\}$, the manifold $S^3_r(K)$ has at most two nontrivial prime summands.*

We first state precisely the published surgery results that will be used, and then prove the required group-theoretic estimate.

4 Inputs from surgery theory

Theorem 2 (Fiberedness)**.** *If $K \subset S^3$ admits a positive L-space surgery, then its exterior fibers over S^1, with fiber a Seifert surface of genus $g(K)$.*

This is the fiberedness assertion of [BS22, Theorem 1], which supplies a proof of the result.

Theorem 3 (Hom–Lidman–Zufelt)**.** *Suppose that a nontrivial knot $K \subset S^3$ admits a positive L-space surgery and is not a cable knot, where nontrivial torus knots are counted as cables of the unknot. If $S^3_r(K)$ is reducible, then*

$$r = 2g(K) - 1.$$

We use the non-cabled formulation stated in the abstract of [HLZ15]. Their Theorem 1.3 establishes the hyperbolic case, and their Section 1 explains the reduction from the non-cabled case using the established torus- and satellite-knot results.

Theorem 4 (Howie)**.** *If a manifold obtained by Dehn surgery on a knot in S^3 has more than two nontrivial prime summands, then it has exactly three. Its prime decomposition has the form*

$$L_1 \# L_2 \# Z,$$

where L_1, L_2 are nontrivial lens spaces and $Z \ncong S^3$ is an integer homology sphere.

This is [How02, Corollary 5.3], including the consequences of the results of Valdez-Sánchez and Sayari incorporated in that corollary.

We will also use the Poincaré theorem: a closed simply connected 3-manifold is homeomorphic to S^3; see [MT07].

> **Reviewer 1**
>
> Not citing Perleman is a notable error.

Consequently, a closed orientable 3-manifold with cyclic fundamental group has at most one nontrivial prime summand. Indeed, a decomposition into two non-S^3 factors would, by Seifert–van Kampen and the Poincaré theorem, express its fundamental group as a free product of two nontrivial groups. Such a free product is not cyclic: nonidentity elements from its two factors do not commute, by the reduced-word normal form.

Lemma 5 (The cable case)**.** *Every Dehn surgery on a nontrivial cable knot in S^3, including a nontrivial torus knot, has at most two nontrivial prime summands.*

Proof. It suffices to consider a nonprime surgery. The classical results for cable knots imply that such a surgery is along a cabling slope. For a nontrivial companion, this follows from [Sch90, Corollary 4.4 and the proof of Corollary 4.5]; the torus-knot case is contained in Moser's classification [Mos71].

Thus, for some knot J, we may write

$$K = C_{a,b}(J), \qquad r = ab,$$

where $a \geq 2$, $\gcd(a, b) = 1$, and a is the winding number of the cable. The cable-surgery formula [Gor83, Corollary 7.3] gives

$$S^3_{ab}(C_{a,b}(J)) \cong L \# S^3_{b/a}(J),$$

where L is a lens space with fundamental group of order a. In particular, L is prime by the preceding observation.

Since $a \geq 2$ and $\gcd(a, b) = 1$, the slope b/a is nonintegral. If J is nontrivial, the nonintegral-surgery theorem of Gordon–Luecke [GL87] says that $S^3_{b/a}(J)$ is irreducible, hence prime. If J is the unknot, its exterior has cyclic fundamental group, and the fundamental group of every filling is a quotient of that group. Thus $S^3_{b/a}(J)$ again has at most one nontrivial prime summand. The displayed connected sum therefore has at most two.

> **Reviewer 1**
>
> This proof can be dramatically shortened though the proof seems correct.

□

5 A genus bound for homology sphere summands

For a finitely generated group G, let $\mathrm{rk}(G)$ denote the minimum size of a generating set. This is group-theoretic generating rank, not the rank of the abelianization.

Lemma 6. *Let $K \subset S^3$ be a fibered knot of genus g, and let p/q be a reduced finite slope with $p \neq 0$. Suppose*

$$S^3_{p/q}(K) \cong X \# Z,$$

where $Z \ncong S^3$ is an integer homology sphere. Then

$$|p| \leq g.$$

Proof. Set

$$E = S^3 \setminus \operatorname{int} \nu(K), \qquad Y = S^3_{p/q}(K), \qquad G = \pi_1(E), \qquad \Gamma = \pi_1(Y), \qquad n = |p|.$$

For a group A, write $A' = [A, A]$.

First, the commutator subgroup of Γ has at most $2g$ generators. The fibration of E has fiber a once-punctured genus-g surface F. Its homotopy exact sequence gives

$$1 \longrightarrow \pi_1(F) \longrightarrow G \longrightarrow \mathbb{Z} \longrightarrow 1.$$

The homomorphism to $\mathbb{Z}$ is the abelianization: a meridian maps to a generator, and $H_1(E) \cong \mathbb{Z}$. Hence

$$G' \cong \pi_1(F) \cong F_{2g},$$

where F_{2g} is the free group of rank $2g$.

Seifert–van Kampen identifies the filling homomorphism as the surjection

$$G \twoheadrightarrow G / \langle\langle \mu^p \lambda^q \rangle\rangle \cong \Gamma.$$

A surjective homomorphism maps the commutator subgroup onto the commutator subgroup: every commutator in the target is the image of a commutator of lifts. Therefore $G' \twoheadrightarrow \Gamma'$, and

$$\operatorname{rk}(\Gamma') \leq 2g. \tag{1}$$

Next, the homology sphere summand forces at least $2n$ generators. The standard surgery homology calculation gives

$$H_1(Y) \cong \mathbb{Z}/n\mathbb{Z}. \tag{2}$$

Indeed, $H_1(E)$ is generated by the meridian, the longitude is null-homologous, and filling imposes the relation $p[\mu] = 0$.

Because $Y = X \# Z$ and $H_1(Z) = 0$, the connected-sum homology formula gives

$$H_1(X) \cong \mathbb{Z}/n\mathbb{Z}.$$

Let $\widetilde{X} \to X$ be the connected n-sheeted covering corresponding to

$$\ker\big(\pi_1(X) \longrightarrow H_1(X)\big).$$

For $n = 1$, this is simply the identity covering.

Choose the balls used to form $X \# Z$, and write

$$X^\circ = X \setminus \operatorname{int} B^3, \qquad Z^\circ = Z \setminus \operatorname{int} B^3.$$

The inverse image of the ball in X consists of n disjoint balls in $\widetilde{X}$. Attach one copy of Z° to each of the resulting boundary spheres. This constructs a connected n-sheeted covering

$$\widetilde{Y} \longrightarrow Y$$

with

$$\widetilde{Y} \cong \widetilde{X} \# \underbrace{Z \# \cdots \# Z}_{n \text{ copies}}. \tag{3}$$

To identify the covering subgroup, observe that

$$\pi_1(Y) \cong \pi_1(X) * \pi_1(Z).$$

The covering monodromy is abelianization on the first factor and is trivial on the second, since $H_1(Z) = 0$. It is therefore precisely the abelianization

$$\Gamma \longrightarrow H_1(Y) \cong \mathbb{Z}/n\mathbb{Z}.$$

Consequently,

$$\pi_1(\widetilde{Y}) \cong \Gamma'.$$

The group $B = \pi_1(Z)$ is perfect because $H_1(Z) = 0$. It is nontrivial by the Poincaré theorem and the assumption $Z \ncong S^3$. A nontrivial perfect group cannot be generated by one element, since a one-generator group is abelian. Thus

$$\operatorname{rk}(B) \geq 2.$$

Applying Seifert–van Kampen to (3) gives

$$\Gamma' \cong \pi_1(\widetilde{X}) * \underbrace{B * \cdots * B}_{n \text{ factors}}.$$

Grushko's rank theorem states that generating rank is additive under free products of finitely generated groups; see [Sta65].

> **Reviewer 1**
>
> Strange citation that is unneeded.

It follows that

$$\operatorname{rk}(\Gamma') = \operatorname{rk}\big(\pi_1(\widetilde{X})\big) + n \operatorname{rk}(B) \geq 2n. \tag{4}$$

Combining (1) and (4) yields $2n \leq 2g$, and hence $n = |p| \leq g$. □

6 Proof of the main theorem

Proof of Theorem 1. Mirroring changes surgery coefficients by a sign and reverses the orientation of the filled manifold:

$$S^3_{-r}(\overline{K}) \cong -S^3_r(K).$$

The L-space condition is invariant under orientation reversal, as is the number of prime summands. We may therefore assume that K admits a positive L-space surgery; compare [HLZ15, Section 1].

Meridional surgery gives S^3, so assume r is finite. If K is the unknot, its exterior has cyclic fundamental group, and every filling has cyclic fundamental group. The observation following Theorem 4 settles this case. If K is a nontrivial cable, Lemma 5 settles the claim.

We may consequently assume that K is nontrivial and non-cabled.

> **Reviewer 1**
>
> We can assume K is hyperbolic, would be more clear.

Suppose, for a contradiction, that

$$Y = S^3_r(K)$$

has more than two nontrivial prime summands. In particular, Y is reducible. Put $g = g(K)$. By Theorem 3,

$$r = n = 2g - 1.$$

Since K is nontrivial, $g \geq 1$, so $n > 0$.

> **Reviewer 1**
>
> Assumes hyperbolic here

Theorem 4 supplies a decomposition

$$Y \cong L_1 \# L_2 \# Z,$$

where L_1, L_2 are nontrivial lens spaces and $Z \ncong S^3$ is an integer homology sphere. Let

$$a_i = |H_1(L_i)|, \qquad i = 1, 2.$$

Both a_i exceed 1. The surgery homology calculation and the connected-sum homology formula imply

$$n = |H_1(Y)| = a_1 a_2 > 1.$$

Together with $n = 2g - 1$, this shows that $g \geq 2$.

By Theorem 2, K is fibered. Apply Lemma 6 to the integer homology sphere summand Z, taking $X = L_1 \# L_2$. It gives

$$2g - 1 = n \leq g.$$

This is impossible for $g \geq 2$. The supposed decomposition cannot exist, proving the theorem. □

References

[BS22] J. A. Baldwin and S. Sivek, *L-space knots are fibered and strongly quasipositive*, The Open Book Series **5** (2022), no. 1, 81–94.

[Gor83] C. McA. Gordon, *Dehn surgery and satellite knots*, Trans. Amer. Math. Soc. **275** (1983), no. 2, 687–708.

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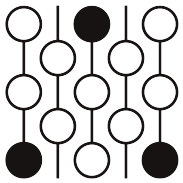
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[Sta65] J. R. Stallings, *A topological proof of Grushko's theorem on free products*, Math. Z. **90** (1965), 1–8.

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October 8–9, 2026



1 Review

Correctness.

- 1: essentially flawless mathematical argument
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Note that points should be docked for wrong statements/ solutions.
Score: 1.

Exposition.

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Score 1.

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
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Score: 0.

I didn't find any errors in the math, though it's convoluted at times.

The argument can be followed without "serious effort", hence a score of 1. That said, the exposition is quite poor. Sometimes it contains weird and unnecessary phrases. Other times it provides wordy or confusing explanations (e.g. "by the surgery homology calculation and the connected-sum homology formula") for somewhat straightforward facts.

The attributions are mostly fine, but there are repeated instances of citing secondary sources.

2 Submission with inline comments

Abstract

Every Dehn surgery on an L-space knot in S^3 has at most two nontrivial prime summands. We prove this using established results about reducing slopes and prime decompositions, together with a group-theoretic genus bound for integer homology sphere summands of surgeries on fibered knots.

> **Reviewer 3**
>
> Abstract is poorly phrased. For instance, it sounds like it's referring to "genus bounds for homology spheres" when presumably it means genus bounds for the knots

3 Statement and conventions

We use the standard ambient-manifold convention: the knots under consideration lie in S^3.

> **Reviewer 3**
>
> Awful opening sentence, knots living in S^3 is hardly a convention.

A rational homology sphere Y is an L-space if

$$\dim_{\mathbb{F}_2} \widehat{HF}(Y;\mathbb{F}_2) = |H_1(Y;\mathbb{Z})|.$$

An L-space knot is a knot admitting a nonmeridional L-space surgery. We allow either sign for this surgery coefficient; when the coefficient can be chosen positive, we explicitly say that the knot admits a positive L-space surgery.

For a knot K, write $S^3_{p/q}(K)$ for Dehn filling along $p\mu + q\lambda$, where μ is a meridian and λ is the Seifert longitude. For finite slopes we take $\gcd(p,q) = 1$ and $q > 0$; the meridional slope is denoted by ∞.

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This is the fiberedness assertion of [BS22, Theorem 1], which supplies a proof of the result.

> **Reviewer 3**
>
> This is weirdly phrased, it mixes the content of the BS theorem with the definition of a fibered knot

Theorem 3 (Hom–Lidman–Zufelt)**.** *Suppose that a nontrivial knot $K \subset S^3$ admits a positive L-space surgery and is not a cable knot, where nontrivial torus knots are counted as cables of the unknot. If $S^3_r(K)$ is reducible, then*

$$r = 2g(K) - 1.$$

> **Reviewer 3**
>
> Again, this is correct but weirdly incompetent: "nontrivial torus knots are counted as cables of the unknot"... What is "are counted as" doing here? Torus knots simply *are* cables of the unknot.

We use the non-cabled formulation stated in the abstract of [HLZ15]. Their Theorem 1.3 establishes the hyperbolic case, and their Section 1 explains the reduction from the non-cabled case using the established torus- and satellite-knot results.

Theorem 4 (Howie)**.** *If a manifold obtained by Dehn surgery on a knot in S^3 has more than two nontrivial prime summands, then it has exactly three. Its prime decomposition has the form*

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where L_1, L_2 are nontrivial lens spaces and $Z \ncong S^3$ is an integer homology sphere.

This is [How02, Corollary 5.3], including the consequences of the results of Valdez-Sánchez and Sayari incorporated in that corollary.

> **Reviewer 3**
>
> missing citations to the original sources

We will also use the Poincaré theorem: a closed simply connected 3-manifold is homeomorphic to S^3; see [MT07].

> **Reviewer 3**
>
> again, maybe Perlman should get a shoutout here? LOL

Consequently, a closed orientable 3-manifold with cyclic fundamental group has at most one nontrivial prime summand. Indeed, a decomposition into two non-S^3 factors would, by Seifert–van Kampen and the Poincaré theorem, express its fundamental group as a free product of two nontrivial groups. Such a free product is not cyclic: nonidentity elements from its two factors do not commute, by the reduced-word normal form.

Lemma 5 (The cable case)**.** *Every Dehn surgery on a nontrivial cable knot in S^3, including a nontrivial torus knot, has at most two nontrivial prime summands.*

> **Reviewer 3**
>
> A poor stylistic choice to single out torus knots.

Proof. It suffices to consider a nonprime surgery.

> **Reviewer 3**
>
> Again, weird opener. Should just start by setting up the notation, as is done in the next paragraph.

The classical results for cable knots imply that such a surgery is along a cabling slope. For a nontrivial companion, this follows from [Sch90, Corollary 4.4 and the proof of Corollary 4.5]; the torus-knot case is contained in Moser's classification [Mos71].

Thus, for some knot J, we may write

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$$S^3_{ab}(C_{a,b}(J)) \cong L \# S^3_{b/a}(J),$$

where L is a lens space with fundamental group of order a. In particular, L is prime by the preceding observation.

> **Reviewer 3**
>
> This is a well-known fact about lens spaces.

Since $a \geq 2$ and $\gcd(a, b) = 1$, the slope b/a is nonintegral. If J is nontrivial, the nonintegral-surgery theorem of Gordon–Luecke [GL87] says that $S^3_{b/a}(J)$ is irreducible, hence prime. If J is the unknot, its exterior has cyclic fundamental group, and the fundamental group of every filling is a quotient of that group. Thus $S^3_{b/a}(J)$ again has at most one nontrivial prime summand. The displayed connected sum therefore has at most two. □

5 A genus bound for homology sphere summands

For a finitely generated group G, let $\mathrm{rk}(G)$ denote the minimum size of a generating set. This is group-theoretic generating rank, not the rank of the abelianization.

> **Reviewer 3**
>
> The above sentence is pure slop.

Lemma 6. *Let $K \subset S^3$ be a fibered knot of genus g, and let p/q be a reduced finite slope with $p \neq 0$. Suppose*

$$S^3_{p/q}(K) \cong X \# Z,$$

where $Z \ncong S^3$ is an integer homology sphere. Then

$$|p| \leq g.$$

Proof. Set

$$E = S^3 \setminus \operatorname{int} \nu(K), \qquad Y = S^3_{p/q}(K), \qquad G = \pi_1(E), \qquad \Gamma = \pi_1(Y), \qquad n = |p|.$$

> **Reviewer 3**
>
> Or, you could just use $|p|$...

For a group A, write $A' = [A, A]$.

First, the commutator subgroup of Γ has at most $2g$ generators. The fibration of E has fiber a once-punctured genus-g surface F. Its homotopy exact sequence gives

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The homomorphism to $\mathbb{Z}$ is the abelianization: a meridian maps to a generator, and $H_1(E) \cong \mathbb{Z}$. Hence

$$G' \cong \pi_1(F) \cong F_{2g},$$

where F_{2g} is the free group of rank $2g$.

Seifert–van Kampen identifies the filling homomorphism as the surjection

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A surjective homomorphism maps the commutator subgroup onto the commutator subgroup: every commutator in the target is the image of a commutator of lifts.

> **Reviewer 3**
>
> verbose and unnecessary

Therefore $G' \twoheadrightarrow \Gamma'$, and

$$\mathrm{rk}(\Gamma') \leq 2g. \tag{1}$$

Next, the homology sphere summand forces at least $2n$ generators. The standard surgery homology calculation gives

$$H_1(Y) \cong \mathbb{Z}/n\mathbb{Z}. \tag{2}$$

Indeed, $H_1(E)$ is generated by the meridian, the longitude is null-homologous, and filling imposes the relation $p[\mu] = 0$.

Because $Y = X \# Z$ and $H_1(Z) = 0$, the connected-sum homology formula gives

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Choose the balls used to form $X \# Z$, and write

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The inverse image of the ball in X consists of n disjoint balls in $\widetilde{X}$. Attach one copy of Z° to each of the resulting boundary spheres. This constructs a connected n-sheeted covering

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To identify the covering subgroup, observe that

$$\pi_1(Y) \cong \pi_1(X) * \pi_1(Z).$$

The covering monodromy is abelianization on the first factor and is trivial on the second, since $H_1(Z) = 0$.

> **Reviewer 3**
>
> Not well justified

It is therefore precisely the abelianization

$$\Gamma \longrightarrow H_1(Y) \cong \mathbb{Z}/n\mathbb{Z}.$$

Consequently,

$$\pi_1(\widetilde{Y}) \cong \Gamma'.$$

The group $B = \pi_1(Z)$ is perfect because $H_1(Z) = 0$. It is nontrivial by the Poincaré theorem and the assumption $Z \ncong S^3$. A nontrivial perfect group cannot be generated by one element, since a one-generator group is abelian. Thus

$$\mathrm{rk}(B) \geq 2.$$

Applying Seifert–van Kampen to (3) gives

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Grushko's rank theorem states that generating rank is additive under free products of finitely generated groups; see [Sta65].

> **Reviewer 3**
>
> Grushko should get a citation

It follows that

$$\operatorname{rk}(\Gamma') = \operatorname{rk}\big(\pi_1(\widetilde{X})\big) + n \operatorname{rk}(B) \geq 2n. \tag{4}$$

Combining (1) and (4) yields $2n \leq 2g$, and hence $n = |p| \leq g$. □

6 Proof of the main theorem

Proof of Theorem 1. Mirroring changes surgery coefficients by a sign and reverses the orientation of the filled manifold:

$$S^3_{-r}(\overline{K}) \cong -S^3_r(K).$$

The L-space condition is invariant under orientation reversal, as is the number of prime summands. We may therefore assume that K admits a positive L-space surgery; compare [HLZ15, Section 1].

Meridional surgery gives S^3, so assume r is finite. If K is the unknot, its exterior has cyclic fundamental group, and every filling has cyclic fundamental group. The observation following Theorem 4 settles this case. If K is a nontrivial cable, Lemma 5 settles the claim.

We may consequently assume that K is nontrivial and non-cabled. Suppose, for a contradiction, that

$$Y = S^3_r(K)$$

has more than two nontrivial prime summands. In particular, Y is reducible. Put $g = g(K)$. By Theorem 3,

$$r = n = 2g - 1.$$

Since K is nontrivial, $g \geq 1$, so $n > 0$.

Theorem 4 supplies a decomposition

$$Y \cong L_1 \# L_2 \# Z,$$

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By Theorem 2, K is fibered. Apply Lemma 6 to the integer homology sphere summand Z, taking $X = L_1 \# L_2$. It gives

$$2g - 1 = n \leq g.$$

This is impossible for $g \geq 2$. The supposed decomposition cannot exist, proving the theorem. □

References

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